INSTRUCTIONS

CIRCUIT BREAKER TRIP CARD
FOR USE WITH VALUTROL*
FULL WAVE REGENERATIVE
DC SCR DRIVES

CIRCUIT BREAKER TRIP CARD

193X539AAG01

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GENERAL ELECTRIC

*TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A.

INSTRUCTIONS

BREAKER TRIP CARD

193X539AAG01

WARNING

DISCONNECT ALL POWER TO THE DRIVE BEFORE REMOVING OR INSERTING A PRINTED CIRCUIT CARD. FAILURE TO DO SO MAY CAUSE INJURY OR DEATH TO PERSONNEL AND DAMAGE TO THE DRIVE AND DRIVEN MACHINERY.

GENERAL

This card is primarily designed for use with a DC–3064R regenerative Valutrol* drive. When properly connected it will initiate a circuit breaker trip to disconnect AC power from the drive if a commutation fault or an A.C. line–to–line fault through the SCRs is detected.

DESCRIPTION

115V AC is applied between AC1T and AC2T to generate approximately 155V DC between ST1 and AC2T.

The circuit breaker shunt trip coil should be rated 120–240V AC/125V DC and connected between ST1 and ST2.

+20V DC is applied from +20V to COM. The TRIP input connects to the corresponding TRIP output point of the Auxiliary Control Card.

When a fault is detected the voltage at TRIP switches to +18V to turn on transistor T12. This generates a gate pulse which is coupled through the pulse transformer to turn on SCR1. When SCR1 turns on the rectified C1 capacitor voltage is applied across the shunt trip coil to open up the circuit breaker. C1 stores energy to provide trip power for faults resulting from A.C. power loss.

CARD TEST

The card can be tested for proper operating as follows:

On the Auxiliary Control Card (193X538A_G01) a jumper wire connects post CFX to post CFY. With power applied to the drive connect CFY to common to simulate a commutation fault. The voltage at TRIP should go positive and immediately cause a circuit breaker trip.

TROUBLESHOOTING

1. Check for proper card operation as described under Card Test.

2. Monitor the voltage at TRIP and at connector posts CT and CFY on the Auxiliary Control Card (193X538A_G01). The voltage at TRIP should only switch positive if:

   a) The CT voltage exceeds +10V (indicating a tray to tray fault), or

   b) The CFY voltage goes to zero (indicating a commutation fault or other trip signal initiation if provided).

3. Excessive voltage transients in the system may cause nuisance tripping. If suspected, make attempts to identify and reduce such transients, for example by adding RC suppression around relay and contactor coils.

GENERAL ELECTRIC 3 4176AA

TITLE PRINTED CIRCUIT DIAGRAM
BREAKER TRIP CARD

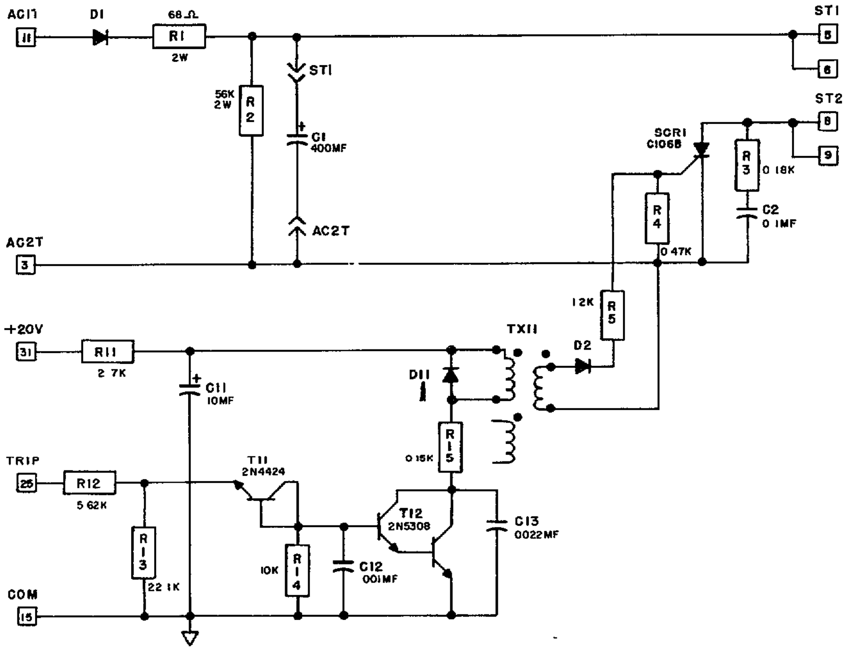
FIRST MADE FOR DG-3064R

193X539AAG01

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING –

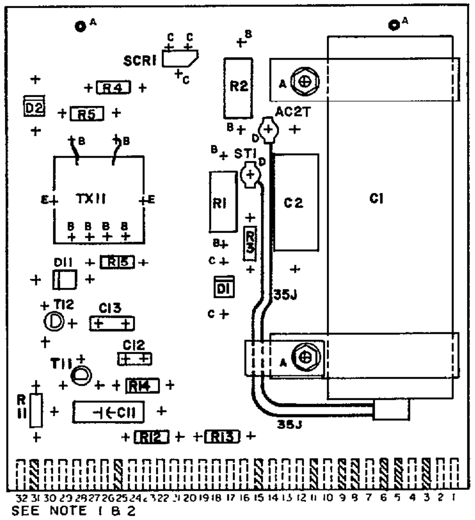
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

SCHEMATIC DIAGRAM



AC11
D1
68 Ω
R1
2W
ST1
5
6
56K 2W
R2
ST1
C1 400MF
AC2T
ST2
8
9
SCR1 C106B
R3 0.18K
C2 0.1MF
R4 0.47K
AC2T
3
12K
R5
+20V
31
R11
2.7K
C11 10MF
TX11
D11
D2
R15
0.15K
TRIP
25
R12
5.62K
T11 2N4424
T12 2N5308
C13 .0022MF
R13
22.1K
10K
R14
C12 .001MF
COM
15

FRONT VIEW SHOWING LOCATION OF COMPONENTS



A
B
SCR1
C C
C
R4
R5
D2
R2
AC2T
B
D
ST1
TX11
E
B B B B
R1
C2
C1
R3
D1
D11
R15
T12
C13
C12
T11
R14
R11
C11
R12
R13
35J
32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

SEE NOTE 1 & 2

HOLE TABULATION

ALL HOLES .040 DIA.
EXCEPT THE HOLES
TABULATED BELOW

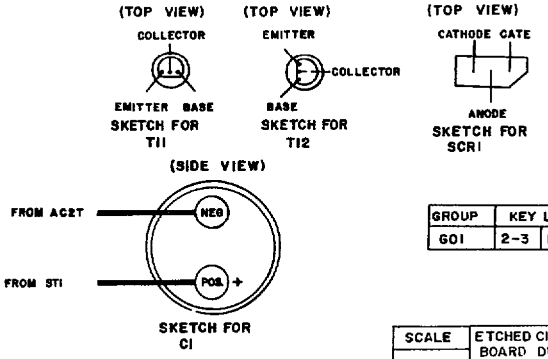
LOC	DIA	QUAN
A	.157	4
B	.052	10
C	.062	5
D	.101	2
E	.128	2

NOTE 1 INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS

2 CROSS HATCHED TABS INDICATE TABS USED

3 CARD SIZE, 5.500 $^{+.000}_{.015}$ X 5.130 $^{+.002}_{.008}$

TRANSISTOR LEAD SKETCHES



(TOP VIEW)
COLLECTOR
EMITTER BASE
SKETCH FOR T11
(TOP VIEW)
EMITTER
COLLECTOR
BASE
SKETCH FOR T12
(TOP VIEW)
CATHODE GATE
ANODE
SKETCH FOR SCR1
(SIDE VIEW)
FROM AC2T
NEG
FROM ST1
POS
SKETCH FOR C1

GROUP	KEY LOCATIONS		
G01	2-3	12-13	19-20

REV NO
0

SCALE	ETCHED CIRCUIT BOARD DWG	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	36B605291AA	3	———

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

5QC(BK)5	5B(S)
AW(BW)	5D(BK)
JA(CD)	5(PCM) 50
	5B(2)
	5(QC)B

MADE BY FEB 28, 1977
AUG 19, 1977
APPROVALS

SPEED VARIATOR
ERIE, PA.

DIV OR DEP
LOCATION

36C764176AA

CONT ON SHT
SH NO

GENERAL ELECTRIC COMPANY – DC MOTOR & GENERATOR DEPARTMENT
SPEED VARIATOR PRODUCTS OPERATION
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC